# H2 Math Summary

Example Questions

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# 1 Systems of Linear Equations

**1.1** Three different blends of coffee beans – Blend A, Blend B and Blend C – are sold at both M Café and I Café. The usual selling prices of the blends are the same at both cafés. The total cost of buying one package each of Blend A, Blend B and Blend C is \$176. During a holiday sale the two cafés offered the following discounts:

| Café | Discounts given for each package | | | Total price after the discount |
|---|---|---|---|---|
| | Blend A | Blend B | Blend C | |
| **M Café** | 15% | 10% | 5% | \$161.80 |
| **I Café** | 7% | 5% | 2% | \$169.56 |

**(i)** Find the usual selling price of each package of Blend A, Blend B and Blend C coffee beans respectively. [3]

**[MI 2025 Prelim P1 Q4]**

| Solution | | Comments |
|---|---|---|
| **(i)**<br>**[3]** | Let the usual prices be \$$a$, \$$b$, \$$c$.<br>$a+b+c=176 \quad ---(1)$<br>$0.85a+0.90b+0.95c=161.80 \quad ---(2)$<br>$0.93a+0.95b+0.98c=169.56 \quad ---(3)$<br>Using GC, $a=32,\ b=44,\ c=100$<br>$\therefore$ Blend A: \$32, Blend B: \$44, Blend C: \$100 | |

**1.2** A function f is defined by $\mathrm{f}(x)=ax^3+bx^2+cx+d$ where $a$, $b$, $c$ and $d$ are constants. The graph of $y=\mathrm{f}(x)$ has a turning point at $x=-1$ and passes through the points $(2, 10)$ and $(-2, -2)$. Given also that $\int_{-2}^{2} \mathrm{f}(x)\,\mathrm{d}x=16$, find the values of $a$, $b$, $c$ and $d$. [4]

**[SAJC 2025 Prelim P1 Q1]**

| **Solution** | | **Comments** |
|---|---|---|
| **[4]** | $\mathrm{f}'(x)=3ax^2+2bx+c$<br>$\mathrm{f}'(-1)=0$<br>$3a-2b+c=0 \quad ---(1)$<br>$\mathrm{f}(2)=10$<br>$8a+4b+2c+d=10 \quad ---(2)$<br>$\mathrm{f}(-2)=-2$<br>$-8a+4b-2c+d=-2 \quad ---(3)$<br>$\int_{-2}^{2} (ax^3+bx^2+cx+d)\,\mathrm{d}x=16$<br>$\left[\frac{a}{4}x^4+\frac{b}{3}x^3+\frac{c}{2}x^2+dx\right]_{-2}^{2}=16$<br>$(4a+\frac{8}{3}b+2c+2d)-(4a-\frac{8}{3}b+2c-2d)=16$<br>$\frac{16}{3}b+4d=16 \quad ---(4)$<br>Using GC, $a=3,\ b=0,\ c=-9,\ d=4$ | |

Two linear equations are independent if and only if neither equation is a linear combination of the other.

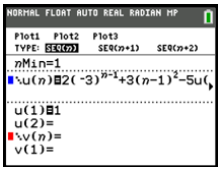
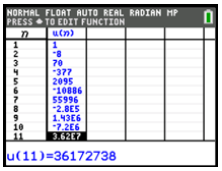
**1.3** A sequence of real numbers, $x_n$, satisfies the recurrence relation

$$x_{n+1} = a(-3)^n + bn^2 + cx_n, \text{ for } n, a, b, c \in \mathbb{Z} \text{ and } n \geqslant 1.$$

Given that $x_1 = 1$, $x_2 = -8$, $x_3 = 70$, and $x_4 = -377$, find the values of $a$, $b$ and $c$. [3]

Hence, find the values of $x_{10}$ and $x_{11}$. [2]

**[HCI 2025 Prelim P1 Q2]**

| **Solution** | | **Comments** |
|---|---|---|
| **[2]** | $n = 1$<br>$x_2 = a(-3)^1 + b(1)^2 + cx_1$<br>$-8 = -3a + b + c \quad ---(1)$<br>$n = 2$<br>$x_3 = a(-3)^2 + b(2)^2 + cx_2$<br>$70 = 9a + 4b - 8c \quad ---(2)$<br>$n = 3$<br>$x_4 = a(-3)^3 + b(3)^2 + cx_3$<br>$-377 = -27a + 9b + 70c \quad ---(3)$<br>Using GC, $a = 2,\ b = 3,\ c = -5$<br>$x_{n+1} = 2(-3)^n + 3n^2 - 5x_n$<br>Using GC,<br>$x_{10} = -7\,210\,868$<br>$x_{11} = 36\,172\,738$<br>  <br><br>  <br> | |

**1.4** The lines $l_1$ and $l_2$ have equations

$$l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix},$$

$$l_2: \mathbf{r} = \begin{pmatrix} 0 \\ 5 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix},$$

where $\lambda$ and $\mu$ are parameters.

**(i)** Show that $l_1$ and $l_2$ are skew lines. [2]

**[SAJC 2025 Prelim P1 Q3]**

| **Solution** | | **Comments** |
|---|---|---|
| **(i)** **[2]** | $1-2\lambda = 2\mu$ $---(1)$<br>$2+\lambda = 5-2\mu$ $---(2)$<br>$2\lambda = -8+3\mu$ $---(3)$<br>Solve (1) and (2): $\lambda = -2,\ \mu = \frac{5}{2}$<br>Sub into (3):<br>$\text{LHS} = 2(-2) = -4$<br>$\text{RHS} = -8+3\left(\frac{5}{2}\right) = -\frac{1}{2}$<br>$\text{LHS} \neq \text{RHS},$ no solution<br>$\therefore l_1$ and $l_2$ do not intersect<br>$\because \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \neq k \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$ for any $k \in \mathbb{R}$<br>$\therefore l_1$ and $l_2$ are not parallel<br>$\therefore l_1$ and $l_2$ are skew (shown) | |

**1.5** A movie theatre supports a charity event by providing discounted movie tickets at the following prices based on age group.

| Age group | Discounted ticket price |
|---|---|
| $\leqslant 12$ years | \$6 |
| 13 to 49 years | \$9 |
| $\geqslant 50$ years (senior citizen) | \$4 |

The event organiser plans to spend \$700 to sponsor 100 participants of which at least 35 of them are senior citizens. Find the possible number of participants for each age group. [4]

**[TJC 2022 Prelim P2 Q1]**

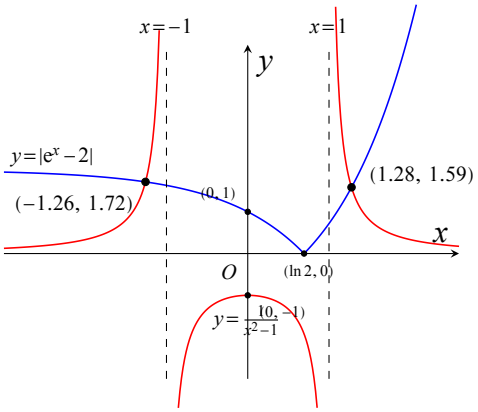
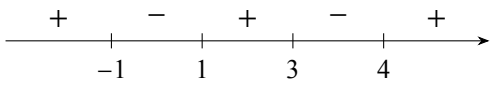
| **Solution** | | **Comments** |
|---|---|---|
| **[4]** | Let $x$, $y$, $z$ be the number of participants aged $\leqslant 12$, 13 to 49 and $\geqslant 50$ years respectively.<br>$\left.\begin{aligned} x+y+z=100 \quad &---(1) \\ 6x+9y+4z=700 \quad &---(2) \end{aligned}\right\} \Rightarrow$ Using GC, $x=\frac{200}{3}-\frac{5}{3}z,\ y=\frac{100}{3}+\frac{2}{3}z$<br>$z \geqslant 35$<br>$\because x \geqslant 0$<br>$\frac{200}{3}-\frac{5}{3}z \geqslant 0$<br>$z \leqslant 40$<br>$\therefore 35 \leqslant z \leqslant 40$<br>$x$, $y$, $z$ are non-negative integers. Using GC table:<br>$z=37 \Rightarrow x=5,\ y=58$<br>$z=40 \Rightarrow x=0,\ y=60$<br>$\therefore (x,y,z)=(5,58,37)$ or $(0,60,40)$ | |

# 2 Inequalities

**2.1** **(a)** Solve the inequality $|e^x - 2| > \dfrac{1}{x^2 - 1}$ using a graphical approach. [3]

**(b)** Without the use of a graphing calculator, solve the inequality $\dfrac{x+2}{x-1} \geqslant \dfrac{2}{x-3}$. [3]

**[RVHS 2026 Prelim P1 Q2]**

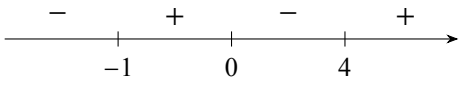
| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** |  <br>Using GC, intersections at $x=-1.2581$, $x=1.2770$<br>$\therefore x<-1.26$ or $-1<x<1$ or $x>1.28$ | |
| **(b)** **[3]** | $\dfrac{x+2}{x-1} - \dfrac{2}{x-3} \geqslant 0$<br>$\dfrac{(x+2)(x-3)-2(x-1)}{(x-1)(x-3)} \geqslant 0$<br>$\dfrac{x^2-3x-4}{(x-1)(x-3)} \geqslant 0$<br>$\dfrac{(x-4)(x+1)}{(x-1)(x-3)} \geqslant 0$<br>  <br>$\therefore x \leqslant -1$ or $1<x<3$ or $x \geqslant 4$ | |

**2.2** Use an algebraic method to solve the inequalities

$$\frac{x(5x+9)}{x^2+2x+5} > x \geqslant 3x-6.$$

[4]

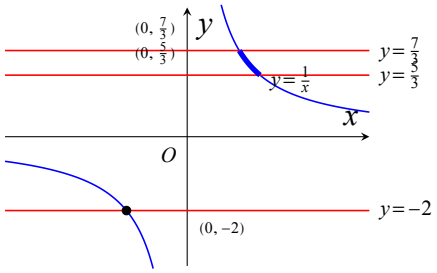
**[ACJC 2026 Prelim P1 Q1]**

| **Solution** | | **Comments** |
|---|---|---|
| **[4]** | $x \geqslant 3x-6$<br>$x \leqslant 3 \quad ---(1)$<br>$x^2+2x+5=(x+1)^2+4>0$ for all real $x$<br>$\frac{x(5x+9)}{x^2+2x+5} > x$<br>$5x^2+9x > x(x^2+2x+5) \quad (\because x^2+2x+5>0)$<br>$x^3-3x^2-4x<0$<br>$x(x-4)(x+1)<0$<br>  <br><br>$x<-1$ or $0<x<4 \quad ---(2)$<br>$(1) \cap (2)$ :<br>$\therefore x<-1$ or $0<x \leqslant 3$ | |

**2.3** **(i)** Solve the inequality $3\,|x^2-4| \leqslant |x+2|$. [4]

**(ii)** Hence, solve $\dfrac{3}{x^2}\,|1-4x^2| \leqslant \left|\dfrac{1+2x}{x}\right|$. [2]

**[SAJC 2025 Prelim P2 Q1]**

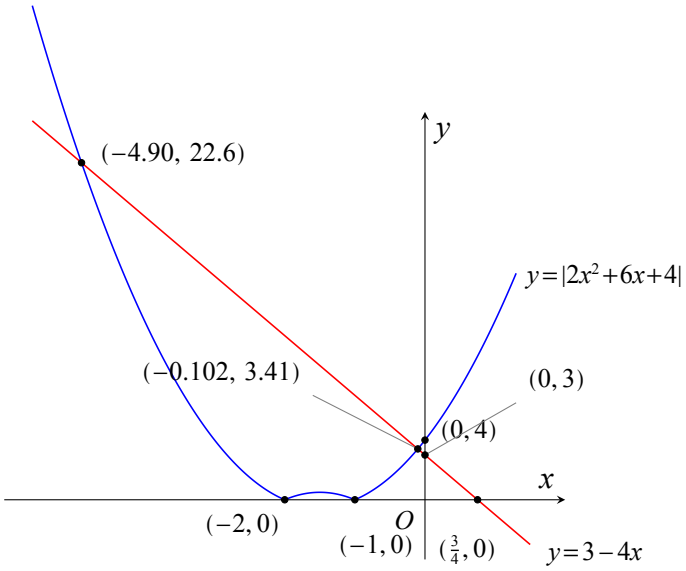
| Solution | | Comments |
|---|---|---|
| **(i)** **[4]** | $3\|x-2\|\|x+2\| \leqslant \|x+2\|$<br>$x=-2$: $0 \leqslant 0$ (true)<br>$x \neq -2$: $\|x+2\| > 0$<br>$3\|x-2\| \leqslant 1$<br>$-\frac{1}{3} \leqslant x-2 \leqslant \frac{1}{3}$<br>$\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$<br>$\therefore x=-2$ or $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$ | |
| **(ii)** **[2]** | $\dfrac{3}{x^2}\,\|1-4x^2\| = 3\left\|\dfrac{1}{x^2}-4\right\| \quad (\because x^2>0)$<br>$\left\|\dfrac{1+2x}{x}\right\| = \left\|\dfrac{1}{x}+2\right\|$<br>$3\left\|\left(\dfrac{1}{x}\right)^2-4\right\| \leqslant \left\|\dfrac{1}{x}+2\right\| \quad (x \to \frac{1}{x})$<br>$\dfrac{1}{x}=-2$ or $\dfrac{5}{3} \leqslant \dfrac{1}{x} \leqslant \dfrac{7}{3}$<br>  <br><br>$\therefore x=-\frac{1}{2}$ or $\frac{3}{7} \leqslant x \leqslant \frac{3}{5}$ | |

**2.4** **(i)** On the same axes, sketch the curves with equations $y=|2x^2+6x+4|$ and $y=3-4x$, indicating any intercepts with the axes and points of intersection. Hence solve the inequality $3-4x<|2x^2+6x+4|$. [4]

**(ii)** Find the exact area bounded by the graphs of $y=3-4x$, $y=|2x^2+6x+4|$, $x=-3$ and $x=-1$. [3]

**[SAJC 2019 Prelim P1 Q2]**

| Solution | Comments |
|---|---|
| **(i)** **[4]** | |




$$3-4x=2x^2+6x+4$$

$$2x^2+10x+1=0$$

$$x=\frac{-5\pm\sqrt{23}}{2}$$

$$\therefore x<\frac{-5-\sqrt{23}}{2} \text{ or } x>\frac{-5+\sqrt{23}}{2}$$

**(ii)** **[3]**

$$\text{Area}=\int_{-3}^{-1}(3-4x-|2x^2+6x+4|)\,dx$$

$$=\int_{-3}^{-2}(3-4x-2x^2-6x-4)\,dx+\int_{-2}^{-1}(3-4x+2x^2+6x+4)\,dx$$

$$=\int_{-3}^{-2}(-2x^2-10x-1)\,dx+\int_{-2}^{-1}(2x^2+2x+7)\,dx$$

$$=\left[-\tfrac{2}{3}x^3-5x^2-x\right]_{-3}^{-2}+\left[\tfrac{2}{3}x^3+x^2+7x\right]_{-2}^{-1}$$

$$=\left(\tfrac{16}{3}-20+2\right)-(18-45+3)+\left(-\tfrac{2}{3}+1-7\right)-\left(-\tfrac{16}{3}+4-14\right)$$

$$=20$$

# 3 Graphing Techniques

**3.1** A curve $C$ has equation $y=\dfrac{x^2-4x+1}{x+1}$, $x\neq -1$.

**(a)** Prove, using an algebraic method, that $y=\dfrac{x^2-4x+1}{x+1}$ cannot lie between two certain values, to be determined in exact form. [3]

**(b)** Sketch $C$, stating the equations of any asymptotes, and the coordinates of any turning points and any axial intercepts. [4]

**[VJC 2025 Promo Q9(a), (b)]**

**(a)** **[3]**

$$y=\frac{x^2-4x+1}{x+1}$$

$$yx+y=x^2-4x+1$$

$$x^2+(-y-4)x+(1-y)=0$$

No real solution for $x$: $(-y-4)^2-4(1-y)<0$

$$y^2+8y+16-4+4y<0$$

$$y^2+12y+12<0$$

$$(y+6)^2-24<0$$

$$\left(y+6-\sqrt{24}\right)\left(y+6+\sqrt{24}\right)<0$$

$$-6-2\sqrt{6}<y<-6+2\sqrt{6}$$

$\therefore$ $y$ cannot lie between $-6-2\sqrt{6}$ and $-6+2\sqrt{6}$

**(b)** **[4]**

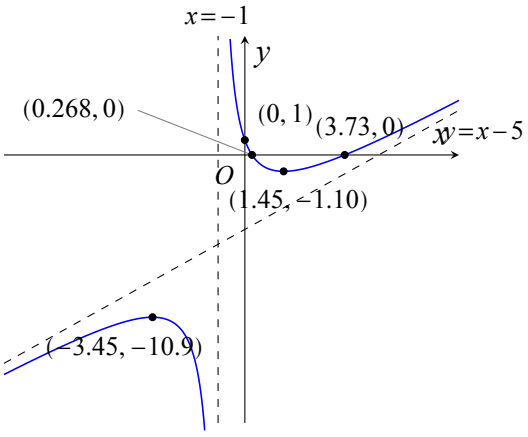
$$y=x-5+\frac{6}{x+1}$$

Asymptotes: $x=-1,\ y=x-5$

$x=0$: $y=1$

$y=0$: $x^2-4x+1=0 \Rightarrow x=0.268,\ 3.73$

Turning points: $(1.45, -1.10),\ (-3.45, -10.9)$



**3.2** A curve $C$ has equation $y=ax+b+\dfrac{2b-a}{x+3}$, where $a$ and $b$ are constants such that $a>0$, $b\neq\dfrac{1}{2}a$ and $x\neq -3$.

**(a)** If $C$ has no stationary points, use differentiation to find the relationship between $a$ and $b$. [3]

**(b)** It is now given that $b=a$.

**(i)** Find the $x$-coordinates of the turning points. [2]

**(ii)** Hence sketch $C$, stating the equations of any asymptotes and the coordinates of the axial intercepts and turning points. [3]

**(iii)** Verify that the point of intersection of the two asymptotes of $C$ lies on the line $y=kx+3k-2a$.

Hence, using the graph in part (b)(ii), find the range of values of $k$ in terms of $a$, such that the equation $x(k-a)+3k-2a-b-\dfrac{2b-a}{x+3}=0$ has real roots. [2]

**[SAJC 2025 Promo Q7]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** | $\dfrac{dy}{dx}=a-\dfrac{2b-a}{(x+3)^2}$<br>$\dfrac{dy}{dx}=0:\quad a-\dfrac{2b-a}{(x+3)^2}=0$<br>$(x+3)^2=\dfrac{2b-a}{a}\qquad ---(1)$<br>No stationary points $\Rightarrow \dfrac{2b-a}{a}<0$<br>Since $a>0,\quad 2b-a<0$<br>$\therefore a>2b$ | |
| **(b)(i)** **[2]** | $b=a:\quad (x+3)^2=\dfrac{2a-a}{a}=1$<br>$x+3=1$ or $-1$<br>$x=-2$ or $-4$ | |

**(b)(ii)**
**[3]**

$$y = ax + a + \frac{a}{x+3}$$

Asymptotes: $x = -3,\ y = ax + a$

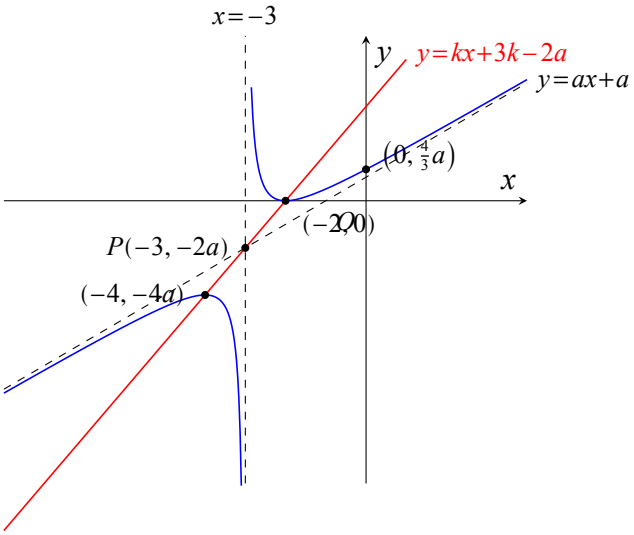
$x = -2 : y = 0; \quad x = -4 : y = -4a$

Turning points: $(-2, 0),\ (-4, -4a)$

$$x = 0 : y = \frac{4}{3}a$$

$$y = 0 : ax + a + \frac{a}{x+3} = 0 \Rightarrow (x+1)(x+3) + 1 = 0$$

$$x^2 + 4x + 4 = 0 \Rightarrow (x+2)^2 = 0 \Rightarrow x = -2$$




**(b)(iii)**
**[2]**

$x = -3$ and $y = ax + a : \quad y = -3a + a = -2a$

Intersection of asymptotes: $P(-3, -2a)$

$x = -3 : \quad \text{RHS} = k(-3) + 3k - 2a = -2a = \text{LHS}$

$\therefore P$ lies on the line $y = kx + 3k - 2a$

$$x(k-a) + 3k - 2a - b - \frac{2b-a}{x+3} = 0$$

$$kx + 3k - 2a = ax + b + \frac{2b-a}{x+3}$$

Line through $P$ meets $C \Rightarrow k > a$

**3.3** A curve $C$ has equation $\mathrm{f}(x)=\dfrac{3x^2+6x+k}{x+2}$, where $x>-2$ and $k$ is a non-zero real constant.

It is given that the function f is strictly increasing.

**(a)** Find the range of values of $k$. [3]

**(b)** Sketch the curve $C$, giving the coordinates of any points where $C$ crosses the $x$- and $y$-axes and the equations of any asymptotes. [3]

**(c)** By adding a suitable curve to the graph of the curve $C$ in (b), deduce the number of distinct real root(s) of the equation $(3x^2+6x+k)^2=(x^2+1)\,(9x^2+36x+36)$. [3]

**[DHS 2025 Promo Q8]**

| **Solution** | | **Comments** |
|---|---|---|
| **(a)** **[3]** | $\mathrm{f}(x)=\dfrac{3x^2+6x+k}{x+2}=3x+\dfrac{k}{x+2}$ | |
| | $\mathrm{f}'(x)=3-\dfrac{k}{(x+2)^2}=\dfrac{3x^2+12x+12-k}{(x+2)^2}$ | |
| | f strictly increasing $\Rightarrow$ $\mathrm{f}'(x)>0$ | |
| | $3x^2+12x+12-k>0$ $\quad\because (x+2)^2>0$ for $x>-2$ | |
| | $3(x+2)^2-k>0$ | |
| | Since $3(x+2)^2-k>-k$ for all $x>-2$, $\quad -k\geqslant 0$ | |
| | $k\neq 0 \Rightarrow k<0$ | |
| **(b)** **[3]** | $y=3x+\dfrac{k}{x+2}$ | |
| | Asymptotes: $y=3x,\ x=-2$ | |
| | $x=0:\ y=\dfrac{k}{2}$ | |
| | $y=0:\ 3x^2+6x+k=0 \Rightarrow x=\dfrac{-6\pm\sqrt{36-12k}}{6}=-1\pm\dfrac{\sqrt{9-3k}}{3}$ | |
| | $x>-2 \Rightarrow x=-1+\dfrac{\sqrt{9-3k}}{3}$ | |

**(c)**
**[3]**

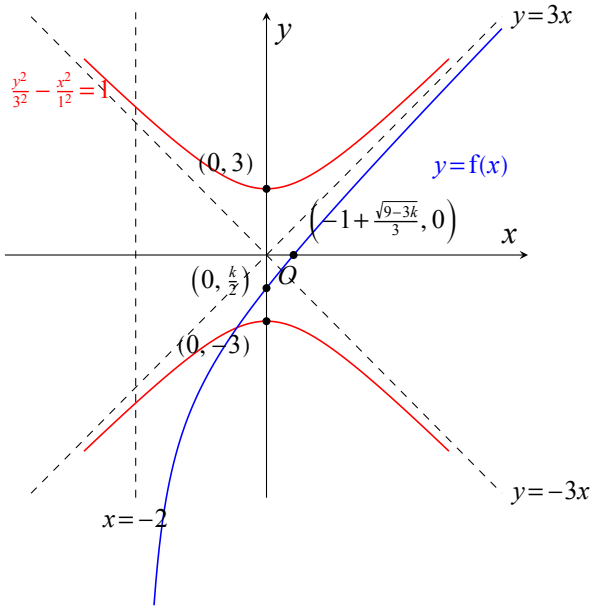
$$(3x^2+6x+k)^2=(x^2+1)\,(9x^2+36x+36) \quad ---(*)$$

$$(3x^2+6x+k)^2=9\,(x^2+1)\,(x+2)^2$$

$$\left(\frac{3x^2+6x+k}{x+2}\right)^2-9x^2=9$$

Draw the hyperbola $\dfrac{y^2}{3^2}-\dfrac{x^2}{1^2}=1$

Asymptotes $y=\pm 3x$, vertices $(0,\pm 3)$




$y=\mathrm{f}(x)$ and $\dfrac{y^2}{3^2}-\dfrac{x^2}{1^2}=1$ have one intersection point

$\therefore$ $(*)$ has one real root

**3.4** The curve $C$ is defined by the parametric equations $x=6t-5$ and $y=2t^2+1$, where $t \geqslant \frac{5}{6}$.

**(a)** Sketch the curve $C$, labelling the exact coordinates of any axial intercept(s). [1]

**(b)** Find the equation of the normal to $C$ at the point $P$ where $t=1$. [2]

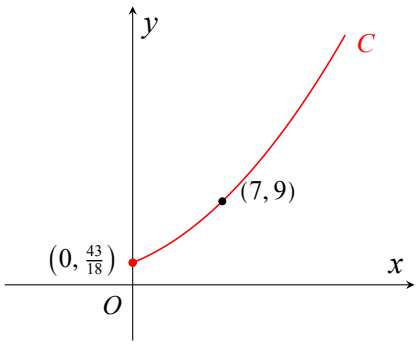
**(c)** Find the acute angle between the normal to $C$ at the point $P$ where $t=1$ and the tangent to $C$ at the point $M$ where $t=3$. [2]

**(d)** The curve $C$ is translated 5 units in the positive $x$-direction and translated 3 units in the negative $y$-direction, to form the curve $D$. Find the equation of $D$ in parametric form. [2]

The curve $E$ is defined by the parametric equations $x=7u$ and $y=\frac{9}{u}$, where $u \neq 0$.

**(e)** Show that at the point of intersection of the curves $C$ and $E$, $6t^3-5t^2+3t-34=0$. Deduce that there is only one point of intersection and find the coordinates of this point. [4]

**[RVHS 2025 Prelim P1 Q12]**

| **Solution** | | **Comments** |
|---|---|---|
| **(a)**<br>**[1]** | $t=\frac{5}{6}$: $x=6\left(\frac{5}{6}\right)-5=0$, $y=2\left(\frac{5}{6}\right)^2+1=\frac{43}{18}$<br>Endpoint ($y$-intercept): $\left(0, \frac{43}{18}\right)$<br>  <br> | |

**(b)** **[2]**

$$\frac{dx}{dt}=6, \quad \frac{dy}{dt}=4t$$

$$\frac{dy}{dx}=\frac{4t}{6}=\frac{2}{3}t$$

$t=1$: $\quad \frac{dy}{dx}=\frac{2}{3} \Rightarrow$ gradient of normal $=-\frac{3}{2}$

$t=1$: $\quad x=1,\ y=3$

$$y-3=-\frac{3}{2}(x-1)$$

$$y=-\frac{3}{2}x+\frac{9}{2}$$

**(c)** **[2]**

$t=3$: $\quad$ gradient of tangent $=\frac{2}{3}(3)=2$

$$\theta_1+\theta_2=\tan^{-1}(2)-\tan^{-1}\left(-\frac{3}{2}\right)=2.089942 \text{ or } 119.7^\circ$$

$$\text{Acute angle}=\pi-2.089942 \text{ or } 180^\circ-119.7^\circ$$

$$=1.05 \text{ or } 60.3^\circ$$

**(d)** **[2]**

$$x-5=6t-5 \Rightarrow x=6t$$

$$y+3=2t^2+1 \Rightarrow y=2t^2-2$$

$D$: $\quad x=6t,\ y=2t^2-2$

**(e)** **[4]**

$$6t-5=7u \Rightarrow u=\frac{6t-5}{7} \quad ---(1)$$

$$2t^2+1=\frac{9}{u} \quad ---(2)$$

Sub (1) into (2): $2t^2+1=\frac{63}{6t-5}$

$$(2t^2+1)(6t-5)=63$$

$$12t^3-10t^2+6t-68=0$$

$$6t^3-5t^2+3t-34=0 \quad \text{(shown)}$$

$$(t-2)(6t^2+7t+17)=0$$

$6t^2+7t+17$: $b^2-4ac=7^2-4(6)(17)=-359<0 \Rightarrow$ no real roots

$\therefore t=2$ only $\Rightarrow$ only one point of intersection

$t=2$: $\quad x=6(2)-5=7,\ y=2(2)^2+1=9$

Point of intersection: $(7, 9)$

**3.5** The parametric equations of the curve $C$ are

$$x = \tan\theta - 1 \quad \text{and} \quad y = 2\sec\theta, \quad \text{where } -\frac{\pi}{2} < \theta < \frac{\pi}{2}.$$

**(a)** Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of $\theta$, leaving your answer in a single trigonometric function. [2]

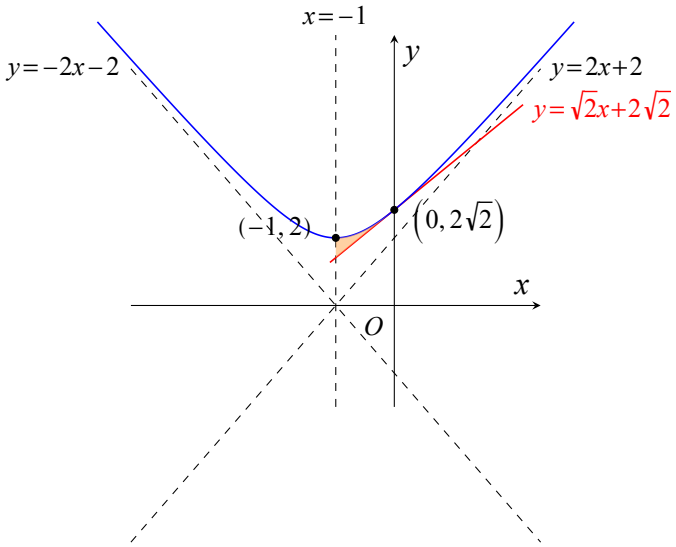
**(b)** Find the equation of the tangent to $C$ at the point where it cuts the $y$-axis, leaving your answer in exact form. [3]

**(c)** Find the Cartesian equation of $C$. [1]

**(d)** Find the equations of the asymptotes and state the coordinates of any turning point(s). [3]

**(e)** The region bounded by the tangent found in part (b), the curve $C$ and the line $x = -1$ is rotated completely about the $x$-axis. Calculate the volume obtained. [3]

**[HCI 2024 Promo Q12]**

| Solution | | Comments |
|---|---|---|
| **(a)** **[2]** | $\dfrac{dy}{dx} = \dfrac{dy}{d\theta} \div \dfrac{dx}{d\theta} = \dfrac{2\sec\theta\tan\theta}{\sec^2\theta} = 2\sin\theta$ | |
| **(b)** **[3]** | $x=0$: $\tan\theta - 1 = 0 \Rightarrow \tan\theta = 1$<br>$\theta = \dfrac{\pi}{4}$ since $-\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2}$<br>$\theta = \dfrac{\pi}{4}$: $y = \dfrac{2}{\cos\frac{\pi}{4}} = 2\sqrt{2}$, $\dfrac{dy}{dx} = 2\sin\dfrac{\pi}{4} = \sqrt{2}$<br>Tangent: $y = \sqrt{2}x + 2\sqrt{2}$ | |
| **(c)** **[1]** | $\tan^2\theta + 1 = \sec^2\theta$<br>$(x+1)^2 + 1 = \left(\dfrac{y}{2}\right)^2$<br>$-\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2} \Rightarrow 0 < \cos\theta \leqslant 1 \Rightarrow \sec\theta \geqslant 1$<br>$\therefore (x+1)^2 + 1 = \left(\dfrac{y}{2}\right)^2, \quad y \geqslant 2$ | |
| **(d)** **[3]** | Asymptotes: $\left(\dfrac{y}{2}\right)^2 = (x+1)^2$<br>$\dfrac{y}{2} = \pm(x+1)$<br>$y = 2x+2$ or $y = -2x-2$<br>Centre $(-1, 0)$, $y \geqslant 2 \Rightarrow$ turning point $(-1, 2)$<br>  <br> | |
| **(e)** **[3]** | $\dfrac{y^2}{4} = (x+1)^2 + 1$<br>$y^2 = 4(x+1)^2 + 4$<br>Volume $= \pi\displaystyle\int_{-1}^{0} \left[4\left((x+1)^2+1\right) - \left(\sqrt{2}x+2\sqrt{2}\right)^2\right] dx$<br>$= 2.09$ units$^3$ | |

## Common Question Types

- Prove algebraically that $y$ cannot lie between two values (range of values $y$ cannot take) 3.1
- Sketch a rational curve, stating asymptotes, turning points and axial intercepts 3.1, 3.2, 3.3
- Condition for no stationary points, using differentiation 3.2
- Find the $x$-coordinates of the turning points 3.2
- Line through the intersection of the asymptotes: range of values of $k$ for real roots 3.2
- Range of values of a constant for a strictly increasing function 3.3
- Add a suitable curve (conic) to the graph to deduce the number of real roots 3.3
- Sketch a parametric curve with an endpoint and axial intercepts 3.4
- Tangent or normal to a parametric curve; acute angle between two lines 3.4, 3.5
- Translate a curve and give its parametric equations 3.4
- Intersection of two parametric curves: show an equation, deduce one point of intersection 3.4
- $\frac{\mathrm{d}y}{\mathrm{d}x}$ of a parametric curve in terms of the parameter 3.5
- Cartesian equation of a parametric curve, then its oblique asymptotes and turning point 3.5
- Volume of revolution of a region bounded by the curve and a tangent 3.5

# 4 Graphing Transformations

**4.1** A curve with equation $y=\mathrm{h}(x)$ undergoes the following sequence of transformations.

**Step 1**: A translation of 2 units in the positive direction of the $x$-axis.

**Step 2**: A scaling parallel to the $x$-axis by factor of 3.

**Step 3**: A reflection in the $y$-axis.

The resultant curve has equation $y=-\dfrac{x^2}{6x+45}$. Find $\mathrm{h}(x)$. [3]

**[NJC 2024 Promo Q4(b)]**

**(b)** **[3]**

$$y = -\frac{x^2}{6x+45}$$

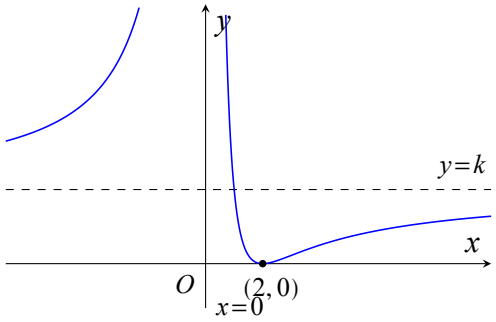
$$(x \to -x): \quad y = -\frac{(-x)^2}{6(-x)+45} = -\frac{x^2}{-6x+45}$$

$$(x \to 3x): \quad y = -\frac{(3x)^2}{-6(3x)+45} = -\frac{9x^2}{-18x+45} = -\frac{x^2}{-2x+5}$$

$$(x \to x+2): \quad y = -\frac{(x+2)^2}{-2(x+2)+5} = -\frac{(x+2)^2}{-2x+1}$$

$$\mathrm{h}(x) = \frac{(x+2)^2}{2x-1}$$

**4.2** The diagram below shows the graph of $y=\mathrm{f}(2x)$. The graph has a turning point at $(2,0)$, and asymptotes with equations $x=0$ and $y=k$.




**(a)** State a single transformation that will transform the graph of $y=\mathrm{f}(2x)$ onto the graph of $y=\mathrm{f}(2x+4)$. Hence sketch the graph of $y=\mathrm{f}(2x+4)$. [3]

On separate clearly labelled diagrams, sketch the graphs of

**(b)** $y=\dfrac{1}{\mathrm{f}(2x)}$, [3]

**(c)** $y=-\mathrm{f}(|x|)$. [3]

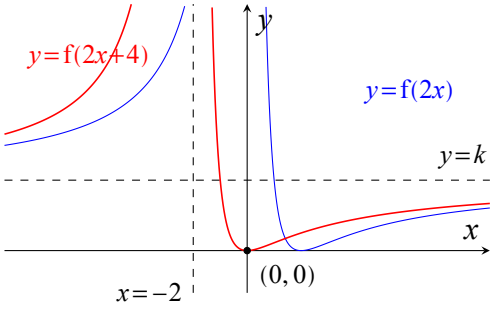
**[EJC 2025 Prelim P1 Q4]**

**(a)** **[3]**

$$f(2x+4) = f(2(x+2)) \qquad (x \to x+2)$$

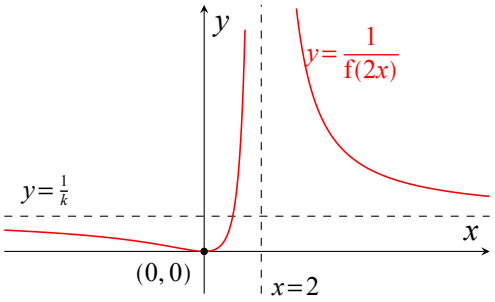
Translation of 2 units in the negative $x$-direction

$$(2, 0) \to (0, 0), \qquad x = 0 \to x = -2$$




**(b)** **[3]**

$$(2, 0) \to x = 2, \qquad x = 0 \to (0, 0), \qquad y = k \to y = \frac{1}{k}$$

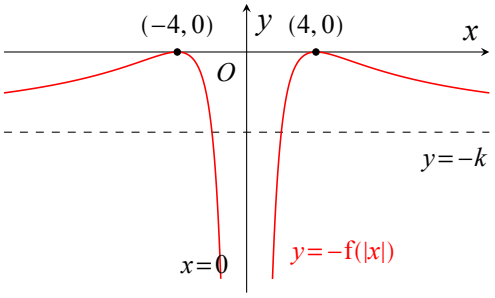



**(c)** **[3]**

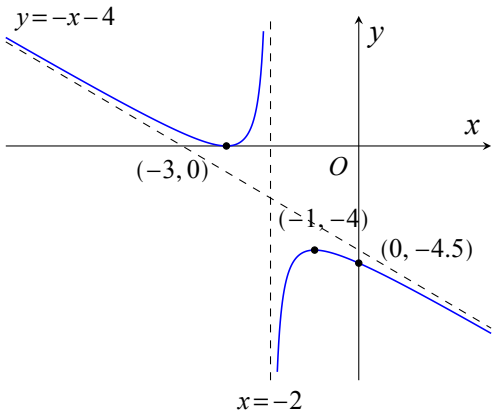
$$f(2x) \xrightarrow{x \to \frac{x}{2}} f(x) \xrightarrow{x \to |x|} f(|x|) \xrightarrow{y \to -y} -f(|x|)$$

$$(2, 0) \to (4, 0) \to (\pm 4, 0) \to (\pm 4, 0)$$

$$y = k \to y = -k, \qquad x = 0$$



**4.3**




The diagram shows the curve $y = \mathrm{f}(x)$. The curve crosses the $x$-axis at its minimum point $(-3, 0)$. It has a maximum turning point $(-1, -4)$ and crosses the $y$-axis at the point $(0, -4.5)$. The asymptotes of the curve are $x = -2$ and $y = -x - 4$.

On separate diagrams, sketch the graphs of

**(a)** $y = \mathrm{f}'(x)$, [3]

**(b)** $y = \dfrac{1}{\mathrm{f}(x)}$, [3]

**(c)** $y = |\mathrm{f}(x)|$, [3]

indicating the equations of any asymptotes, turning points and intersections with the axes.

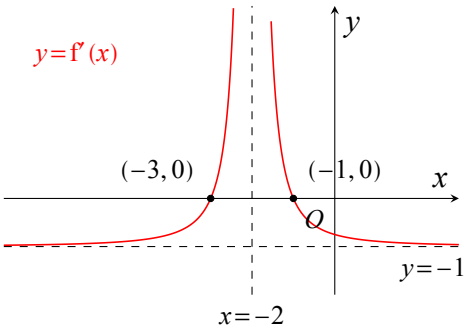
**[CJC 2025 Promo Q8]**

**(a)** **[3]**

Turning points $x=-3,\ -1 \Rightarrow (-3,0),\ (-1,0)$

$y=-x-4 \Rightarrow y=-1$

$x=-2 \Rightarrow x=-2$




**(b)** **[3]**

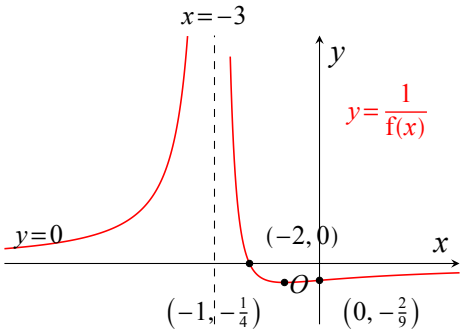
$(-3,0) \Rightarrow x=-3$

$x=-2 \Rightarrow (-2,0)$

$(-1,-4) \Rightarrow \left(-1,-\frac{1}{4}\right)$

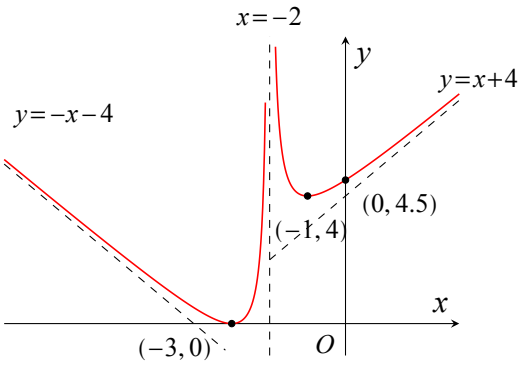
$(0,-4.5) \Rightarrow \left(0,-\frac{2}{9}\right)$

$y=-x-4 \Rightarrow y=0$



**(c)** **[3]**

$$(-1, -4) \Rightarrow (-1, 4)$$
$$(0, -4.5) \Rightarrow (0, 4.5)$$
$$y = -x - 4 \ (x > -2) \Rightarrow y = x + 4$$



**4.4** Express $\frac{9x+5}{3x+2}$ in the form $a-\frac{b}{3x+2}$ where $a$ and $b$ are positive constants.

Hence, describe a sequence of transformations which transform the graph of $y=\frac{9x+5}{3x+2}$ to the graph of $y=\frac{1}{x}$. [5]

**[RVHS 2026 Prelim P1 Q10(b)]**

**(b)** **[5]**

$$\frac{9x+5}{3x+2}=3-\frac{1}{3x+2} \qquad (a=3,\ b=1)$$

$$y=3-\frac{1}{3x+2} \xrightarrow{y\to y+3} y=-\frac{1}{3x+2}$$

$$\xrightarrow{y\to -y} y=\frac{1}{3x+2}$$

$$\xrightarrow{x\to \frac{x}{3}} y=\frac{1}{x+2}$$

$$\xrightarrow{x\to x-2} y=\frac{1}{x}$$

1. Translate 3 units in the negative $y$-direction
2. Reflect about the $x$-axis
3. Scale by factor 3 parallel to the $x$-axis
4. Translate 2 units in the positive $x$-direction

# 5 Functions

**5.1** The function f is defined by

$$\mathrm{f}: x \to \left|\frac{2x+4}{3-x}\right|, \qquad x \in \mathbb{R}, \qquad x \neq 3.$$

**(a)** Sketch the graph of $y = \mathrm{f}(x)$, giving the equations of any asymptotes and the coordinates of the points where the curve meets the axes. [3]

**(b)** Hence state the range of f. [1]

**(c)** Explain why the function $\mathrm{f}^2$ does not exist. [1]

The domain of f is further restricted to $x \leqslant a$, where $a$ is a constant.

**(d)** State the greatest value of $a$ such that the function $\mathrm{f}^{-1}$ exists. [1]

**(e)** Hence find $\mathrm{f}^{-1}(x)$ and state its domain. [4]

**[2023 A-Level P1 Q7]**

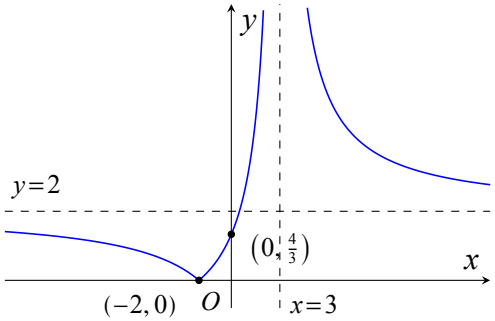
**(a)** **[3]**

$$y=\left|\frac{2x+4}{3-x}\right|=\left|-2+\frac{10}{3-x}\right|$$

Asymptotes: $y=|-2|=2, \quad x=3$

$$x=0: \quad y=\left|\frac{4}{3}\right|=\frac{4}{3}$$

$$y=0: \quad x=-2$$




**(b)** **[1]**

$$\mathrm{R_f}=[0,\infty)$$

**(c)** **[1]**

$$\mathrm{R_f}=[0,\infty), \quad \mathrm{D_f}=(-\infty,3)\cup(3,\infty)$$

$$3\in\mathrm{R_f},\ 3\notin\mathrm{D_f} \Rightarrow \mathrm{R_f}\not\subseteq\mathrm{D_f}$$

$\therefore$ $\mathrm{f}^2$ does not exist

**(d)** **[1]**

$$a=-2$$

**(e)** **[4]**

$$x\leqslant -2: \quad \mathrm{f}(x)=\left|\frac{2x+4}{3-x}\right|=-\frac{2x+4}{3-x}$$

$$y=\mathrm{f}(x)$$

$$-3y+xy=2x+4$$

$$x(y-2)=4+3y$$

$$x=\frac{4+3y}{y-2}$$

$$\mathrm{f}^{-1}(x)=\frac{4+3x}{x-2}, \qquad \mathrm{D_{f^{-1}}}=[0,2)$$

**5.2** The function f is defined by

$$\mathrm{f}: x \mapsto \frac{ax-3}{5x-4}, \text{ for } x \in \mathbb{R}, \ x \neq \frac{4}{5}, \text{ and } a \in \mathbb{R}.$$

**(a)** Find the range of f in terms of $a$. [1]

The function f is such that $\mathrm{f}(x) = \mathrm{f}^{-1}(x)$ for all $x$ in the domain of f.

**(b)** Find the value of $a$. [2]

**(c)** Hence find $\mathrm{f}^{2025}(2)$. [2]

Another function g is defined by

$$\mathrm{g}: x \mapsto \left(x - \frac{1}{5}\right)^2 + \frac{2}{5}, \text{ for } x \in \mathbb{R}, \ 0 < x < \frac{3}{5}.$$

**(d)** Find the range of fg. [2]

**[HCI 2025 Prelim P1 Q8]**

**(a)** **[1]**

Horizontal asymptote: $y=\frac{a}{5}$

$$\mathrm{R_f}=\mathbb{R}\setminus\left\{\frac{a}{5}\right\}$$

**(b)** **[2]**

$$y=\frac{ax-3}{5x-4}$$
$$5xy-4y=ax-3$$
$$5xy-ax=4y-3$$
$$x=\frac{4y-3}{5y-a}$$
$$\mathrm{f}^{-1}(x)=\frac{4x-3}{5x-a},\quad x\in\mathbb{R},\ x\neq\frac{a}{5}$$
$$\mathrm{f}(x)=\mathrm{f}^{-1}(x)\ \Rightarrow\ a=4$$

**(c)** **[2]**

$$\mathrm{f}^{-1}(x)=\mathrm{f}(x)\ \Rightarrow\ \mathrm{f}^2(x)=\mathrm{ff}^{-1}(x)=x$$
$$\mathrm{f}^{2025}(2)=\mathrm{f}^2\left(\mathrm{f}^{2023}(2)\right)$$
$$=\mathrm{f}^2\left(\mathrm{f}^{2021}(2)\right)$$
$$=\cdots$$
$$=\mathrm{f}(2)$$
$$=\frac{4(2)-3}{5(2)-4}=\frac{5}{6}$$

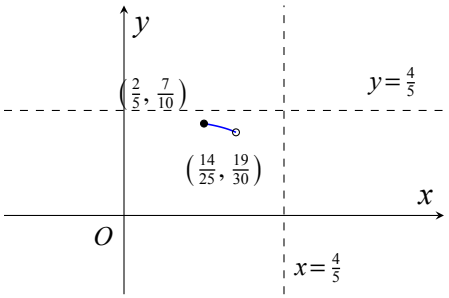
**(d)** **[2]**

$$\mathrm{R_g} = \left[\frac{2}{5}, \frac{14}{25}\right)$$

$$\mathrm{f}(x) = \frac{4x-3}{5x-4}$$

$$\left(0, \frac{3}{5}\right) \xrightarrow{\mathrm{g}} \left[\frac{2}{5}, \frac{14}{25}\right) \xrightarrow{\mathrm{f}} \left(\frac{19}{30}, \frac{7}{10}\right]$$

$$\mathrm{R_{fg}} = \left(\frac{19}{30}, \frac{7}{10}\right]$$



**5.3** The function f is given by

$$\mathrm{f}(x)=\begin{cases}2+\sqrt{2^2-(x-2)^2}, & 0<x\leqslant 4,\\ 2-\sqrt{2^2-(x-6)^2}, & 4<x\leqslant 8.\end{cases}$$

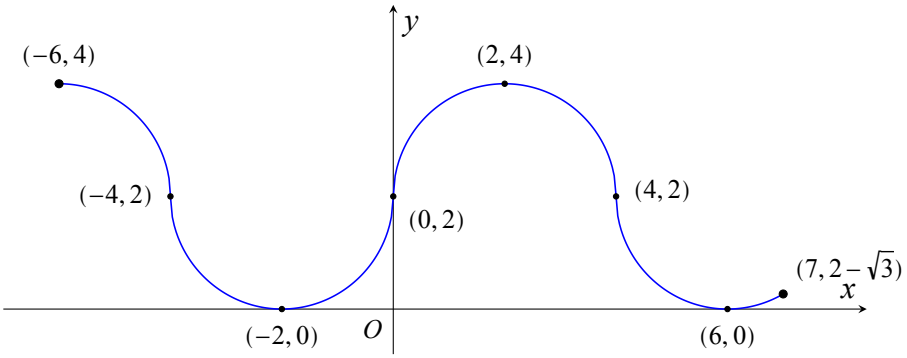
It is given that $\mathrm{f}(x)=\mathrm{f}(x+8)$ for all real values of $x$.

**(a)** On the diagram in the Printed Answer Book, sketch the graph of $y=\mathrm{f}(x)$ for $-6\leqslant x\leqslant 7$, indicating clearly the coordinates of the end points and the points where the graph cuts the axes. [3]

**(b)** Without integrating, write down the exact area of the region bounded by $y=\mathrm{f}(x)$, the line $x=4$, the $x$-axis and the $y$-axis. [1]

**[VJC 2025 Prelim P1 Q5(a), (b)]**

**(a)** **[3]**




**(b)** **[1]**

$$\text{Area} = 2\times 4 + \frac{1}{2}\pi(2^2) = 8 + 2\pi$$

**5.4** The function f is defined by

$$\mathrm{f}(x)=\begin{cases}2-\ln(x+\mathrm{e}) & \text{for } -\mathrm{e}<x<0,\\ \cos\left(\dfrac{\pi}{2}x\right) & \text{for } 0\leqslant x\leqslant 1.\end{cases}$$

**(a)** Given that the inverse function $\mathrm{f}^{-1}$ exists, find the exact value of $m$ if $\mathrm{f}^{-1}(m)=-1$. [2]

Another function g is defined by

$$\mathrm{g}(x)=\sin x, \qquad \text{for } 0\leqslant x\leqslant \frac{\pi}{4}.$$

**(b)** Explain why the composite function fg exists. [2]

**(c)** Find the exact value of $\mathrm{fg}\left(\dfrac{\pi}{6}\right)$. [2]

The domain of f is further restricted to $0\leqslant x\leqslant 1$.

**(d)** By considering $\mathrm{f}^2(1)$ and $\mathrm{f}^3(1)$, deduce the value(s) of $\mathrm{f}^n(1)$ where $n$ is a positive integer. [2]

**[HCI 2024 Promo Q8]**

**(a)** **[2]**

$$\begin{aligned} f^{-1}(m) &= -1 \\ m &= f(-1) \\ &= 2 - \ln(e-1) \end{aligned}$$

**(b)** **[2]**

$$R_g = \left[0, \frac{1}{\sqrt{2}}\right]$$

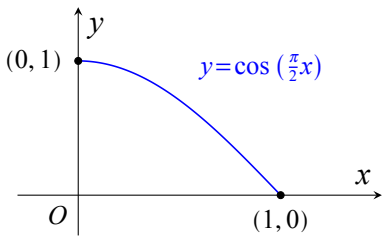
$$D_f = (-e, 1]$$

$$R_g \subseteq D_f \Rightarrow \text{ fg exists}$$

**(c)** **[2]**

$$\begin{aligned} fg\left(\frac{\pi}{6}\right) &= f\left(\frac{1}{2}\right) \\ &= \cos\left(\frac{\pi}{4}\right) \\ &= \frac{1}{\sqrt{2}} \end{aligned}$$

**(d)** **[2]**




$$f^2(1) = f[f(1)] = f(0) = 1$$

$$f^3(1) = f[f^2(1)] = f(1) = 0$$

$$f^n(1) = \begin{cases} 0 & \text{if } n \text{ is odd} \\ 1 & \text{if } n \text{ is even} \end{cases}$$

**5.5** The function f is defined by

$$\mathrm{f}(x) = 1 - ax,\ x \in \mathbb{R},\ \text{where } a \text{ is a real constant.}$$

The function g is defined by

$$\mathrm{g}(x) = \begin{cases} x^2 + 3 & \text{for} \quad 0 \leqslant x \leqslant 2, \\ 7 - x & \text{for} \quad 2 < x \leqslant 7. \end{cases}$$

**(i)** Find the set of possible values of $a$ such that $\mathrm{f}^{-1}$ exist. [1]

**(ii)** If $a = 2$, describe a sequence of transformations that transform the graph of $y = \mathrm{g}(x)$ onto the graph of $y = \mathrm{fg}(x)$. [3]

**(iii)** The function gg is defined by

$$\mathrm{gg}(x) = \begin{cases} \mathrm{h}_1(x) & \text{for} \quad 0 \leqslant x \leqslant 2, \\ \mathrm{h}_2(x) & \text{for} \quad 2 < x < 5, \\ \mathrm{h}_3(x) & \text{for} \quad 5 \leqslant x \leqslant 7. \end{cases}$$

Find $\mathrm{h}_1(x)$, $\mathrm{h}_2(x)$ and $\mathrm{h}_3(x)$. [3]

**[ACJC 2022 Prelim P2 Q1]**

**(i)** **[1]**

$$a \in \mathbb{R} \setminus \{0\}$$

**(ii)** **[3]**

$$y = \mathrm{fg}(x) = 1 - 2\mathrm{g}(x)$$

Reflection of the graph of $y = \mathrm{g}(x)$ in the $x$-axis, followed by scaling by scale factor 2 parallel to the $y$-axis, followed by translation 1 unit in the positive $y$-direction.

**(iii)** **[3]**

$$\mathrm{gg}(x) = \begin{cases} 7 - (x^2 + 3) & \text{for} \quad 0 \leqslant x \leqslant 2 \\ 7 - (7 - x) & \text{for} \quad 2 < x < 5 \\ (7 - x)^2 + 3 & \text{for} \quad 5 \leqslant x \leqslant 7 \end{cases}$$

$$\therefore \mathrm{gg}(x) = \begin{cases} 4 - x^2 & \text{for} \quad 0 \leqslant x \leqslant 2 \\ x & \text{for} \quad 2 < x < 5 \\ x^2 - 14x + 52 & \text{for} \quad 5 \leqslant x \leqslant 7 \end{cases}$$

**5.6** Functions f and g are defined by

$$\mathrm{f}: x \to \frac{4}{(x-4)^2}, \quad x \in \mathbb{R},\ x \neq 4,$$

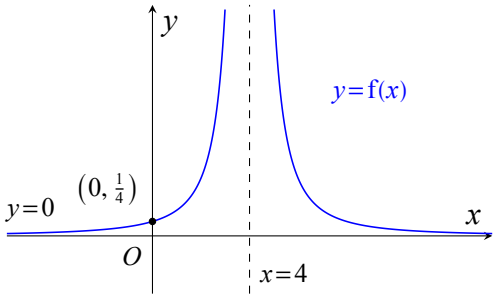
$$\mathrm{g}: x \to \ln\left(1+\frac{1}{x}\right), \quad x \in \mathbb{R},\ x > 0.$$

**(a)** Sketch the graph of $y = \mathrm{f}(x)$, stating the equations of any asymptotes, the coordinates of the points where it crosses the axes and the coordinates of the turning points, if any. [2]

**(b)** Show that gf exists. Hence find the rule, domain and range of gf. [4]

**[CJC 2025 Prelim P2 Q2(a), (b)]**

**(a)** **[2]**




**(b)** **[4]**

$$R_f = (0, \infty)$$

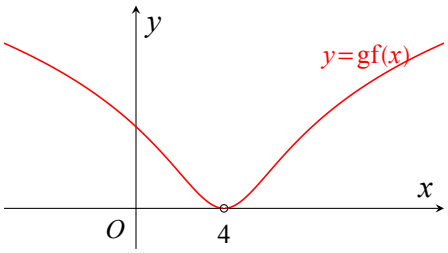
$$D_g = (0, \infty)$$

$$R_f \subseteq D_g \Rightarrow \text{gf exists (shown)}$$

$$\begin{aligned} \text{gf}(x) &= \text{g}\left[\frac{4}{(x-4)^2}\right] \\ &= \ln\left[1+\left(\frac{1}{\frac{4}{(x-4)^2}}\right)\right] \\ &= \ln\left[1+\frac{(x-4)^2}{4}\right] \end{aligned}$$

$$D_{gf} = D_f = (-\infty, 4) \cup (4, \infty)$$

$$D_f = (-\infty, 4) \cup (4, \infty) \xrightarrow{f} R_f = (0, \infty) \xrightarrow{g} R_{gf} = (0, \infty)$$




$$R_{gf} = (0, \infty)$$